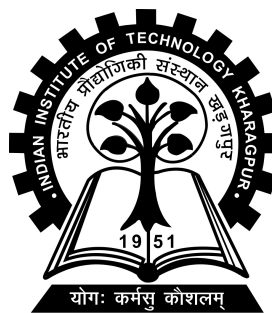


Digitalised Agricultural Value Chain with Integrated Inventory Management and Quality Control Systems

Project-II report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Bachelor of Technology
in
Industrial and Systems Engineering

by
Dronesh Mahendra Magare (21IM30013)

Under the supervision of
Professor Biswajit Mahanty



Department of Industrial and Systems Engineering

Indian Institute of Technology Kharagpur

Spring Semester, 2024-25

March 3, 2025

DECLARATION

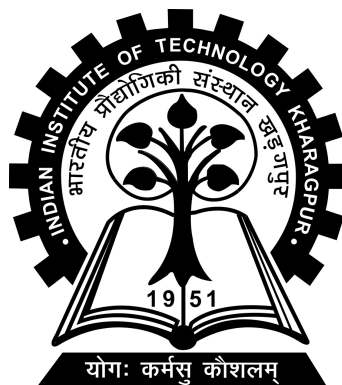
I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: March 3, 2025

Place: Kharagpur

DEPARTMENT OF INDUSTRIAL AND SYSTEMS
ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR - 721302, INDIA



CERTIFICATE

This is to certify that the project report entitled “Digitalised Agricultural Value Chain with Integrated Inventory Management and Quality Control Systems” submitted by Dronesh Mahendra Magare (21IM30013) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Industrial and Systems Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2024-25.

Date: March 3, 2025

Place: Kharagpur

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CHAPTER 1

1. INTRODUCTION

The agricultural supply chain (ASC) in India faces significant challenges, particularly in the green grocery sector. Despite India's vast agricultural resources, including 52 percent arable land (compared to the world average of 11 percent), 15 major climates, and 20 agro-climatic zones, the conventional supply chain based on the mandi system has become a bottleneck in today's economy¹. This system is characterized by price opaqueness, high wastage levels, and reduced returns for producers.

In the green grocery sector, which includes fresh vegetables and fruits, these inefficiencies are particularly acute. Farmers receive only a fraction of the final retail price, while consumers pay inflated costs due to numerous intermediaries and systemic inefficiencies. The lack of modern inventory management systems, quality control mechanisms, and transparent pricing models has led to a disconnect between production and consumption, resulting in significant post-harvest losses estimated at 30-40

This project aims to address these challenges through technological innovation, proposing a comprehensive digital transformation of the green grocery supply chain. The research focuses on developing an asset-light, technology-driven platform that integrates key stakeholders – farmers, sorting centers, distribution centers, processing industries, valorization industries, and markets – while preserving essential elements of the traditional system¹.

The proposed solution introduces several key innovations:

- A Mixed-Integer Linear Programming (MILP) model to optimize the supply chain, minimizing losses and maximizing profits¹.
- A fixed price range model to replace the volatile mandi bidding system.
- An integrated inventory management system with real-time tracking and quality control.
- A transparent supply chain framework that reduces intermediary stages while maintaining traditional stakeholder roles.

2. OBJECTIVE

To develop and implement a comprehensive mathematical model that optimizes the flow of agricultural products through the supply chain, with the primary goal of maximizing overall profit while considering various uncertainties and constraints. Specifically, the project aims to:

1. Minimize total costs associated with transportation, storage, and processing of agricultural products.
2. Maximize revenue by optimizing the distribution of products to different markets based on demand and price fluctuations.
3. Reduce post-harvest losses by efficiently managing the flow of perishable goods through the supply chain.
4. Incorporate uncertainty factors such as weather conditions, market demand variations, and potential supply chain disruptions into the decision-making process.
5. Determine optimal locations for new processing facilities to enhance the efficiency of the supply chain network.
6. Balance the allocation of resources and products across different nodes of the supply chain to ensure overall system efficiency.
7. Provide a decision support tool for stakeholders in the agricultural sector to make informed choices regarding production, distribution, and marketing of agricultural products.

By achieving these objectives, the project aims to contribute to the overall improvement of agricultural supply chain management, leading to increased profitability for farmers and other stakeholders, reduced food waste, and enhanced food security.

CHAPTER 3

3. LITERATURE REVIEW

The agricultural supply chain (ASC) in India has been a subject of extensive research, particularly focusing on its inefficiencies and the potential for technological interventions. Several key issues have been identified in existing literature:

- Price volatility due to mandi-based bidding systems
- Lack of transparency in the supply chain
- Significant post-harvest losses, estimated at 30-40 percent
- Fragmented logistics infrastructure
- Insufficient refrigerated warehousing facilities
- Excessive competition among traders in conventional markets
- Improper handling of post-harvest produce due to lack of training
- Inefficient recording of sales managed by agriculturists
- Intra-day price variations

Recent studies have highlighted the potential of digital technologies in addressing these challenges. AgriTech startups have developed platforms that connect farmers directly with buyers, aiming to eliminate intermediaries and ensure a more equitable distribution of profits. These platforms utilize mobile and web-based applications to increase farmers' incomes and promote transparency in the agricultural supply chain.

However, research suggests that many of these solutions often bypass existing stakeholders, leading to resistance and limited adoption. The conventional agricultural supply chain, based on the mandi system, has become a bottleneck in today's economy, characterized by price opaqueness, high wastage levels, and reduced returns for producers.

CHAPTER 4

4. INDIAN RETAIL FRESH PRODUCE SUPPLY CHAIN

The offline retail supply chain for fruits and vegetables (F&V) remains crucial in India, especially given recent trends highlighted in consumer surveys. This section analyzes the structure, role, and need for optimizing the offline retail supply chain for F&V in India, based on recent consumer data and industry insights.

4.1. Current Landscape of Retail Supply Chains for Fruits and Vegetables in India

India's retail supply chain for fruits and vegetables can be categorized into two main types:

- **Organised Retail Chains:** Large chains such as Reliance Fresh, ABRL More, and ITC Choupal Fresh, which have formal supply chains, advanced procurement methods, quality control systems, and direct partnerships with farmers.
- **Traditional (Unorganised) Retailers:** These include kirana stores, local vendors, roadside vendors, and mandis (wholesale markets). These retailers are informal, rely on intermediaries, and operate without strict quality standards or advanced logistical support.

Data on Offline Retail Supply Chain:

- **Employment:** The traditional retail sector provides jobs to approximately 40 million people in India.
- **Market Share:** Unorganised retail commands about 85% of the total F&V retail market.
- **Frequency of Purchase:** Households typically purchase F&V from offline sources 2-3 times a week.

F & V Supply Chain For Reliance Fresh

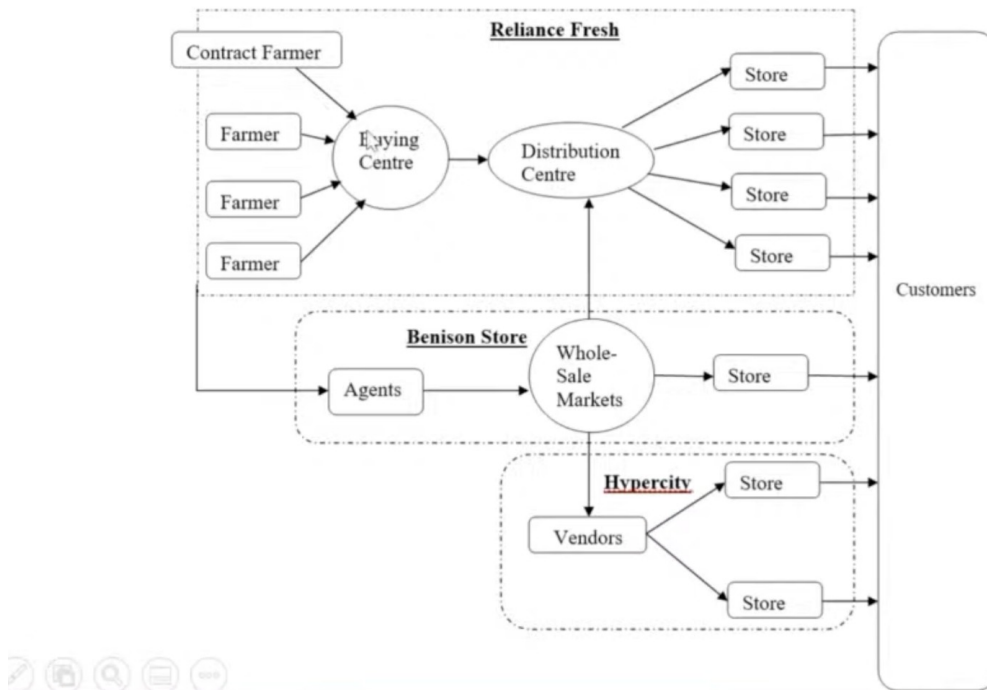


Figure 4.1: Structure of Retail Supply Chains for Fruits and Vegetables in India

4.2. Need for an Effective Offline Retail Supply Chain

- **Quality Control:** Offline vendors allow consumers to inspect produce before purchase, fostering trust in quality.
- **Pricing Sensitivity:** Offline vendors often provide competitive pricing due to lower overheads and flexible market dynamics.
- **Consumer Loyalty and Trust:** Personal interactions and the ability to visually inspect produce create long-established loyalty among consumers for offline markets.

4.3. Challenges in the Current Offline Supply Chain

- **Multi-tiered Structure:** Numerous intermediaries increase handling time, costs, and reduce farmer profitability while raising consumer prices.
- **Post-harvest Losses:** Limited cold chain infrastructure leads to post-harvest losses of 30-40%.

CHAPTER 5

5. MATHEMATICAL MODEL

5.1. Problem Definition

The agricultural supply chain consists of multiple levels:

- Farmers (F): Produce raw crops
- Sorting Centers (SC): Grade and package crops
- Distribution Centers (DC): Store and distribute products
- Processing Industries (PI): Convert low-quality produce into by-products
- Valorization Industries (VI): Process waste into value-added products
- Markets (M): Sell final products

The objective is to optimize product flow while considering:

1. Demand uncertainty
2. Transportation and processing costs
3. Storage limitations
4. Market price fluctuations

5.2. Assumptions

- Facility capacities and transportation costs are fixed.
- Each farmer supplies only one SC.
- Locations of all facilities are predetermined, except for new PIs.
- A time horizon of one year is considered.

5.3. Indices

- $i \in I$ Farmers
- $j \in J$ Sorting Centers
- $d \in D$ Distribution Centers
- $n \in N$ Processing Industries (new and existing)
- $q \in Q$ Quality categories
- $c \in C$ By-product types
- $m \in M$ Markets
- $v \in V$ Valorization Industries
- $t \in T$ Time periods
- $s \in S$ Scenarios (uncertainty consideration)

5.4. Decision Variables

- X_{ij} Quantity transported from i to j
- Y_{jd} Quantity sent from j to d
- Z_{dm} Quantity delivered from d to m
- W_{dn} Quantity shipped from d to n
- U_{nv} Waste sent from n to v

5.5. Objective Function

$$\begin{aligned} \max \sum_{s \in S} p_s \left[\sum_{m \in M} r_m Z_{dm} + \sum_{n \in N} r_n W_{dn} + \sum_{v \in V} r_v U_{nv} \right] - \left[\sum_{i \in I, j \in J} c_{ij} X_{ij} + \sum_{j \in J, d \in D} c_{jd} Y_{jd} \right. \\ \left. + \sum_{d \in D, m \in M} c_{dm} Z_{dm} \right. \\ \left. + \sum_{d \in D, n \in N} c_{dn} W_{dn} + \sum_{n \in N, v \in V} c_{nv} U_{nv} \right] \end{aligned}$$

where:

p_s is the probability of scenario s ,

r_m is the revenue per unit sold at market m ,

r_n is the revenue per unit processed at industry n ,

r_v is the revenue per unit valorized at industry v ,

c_{ij} is the cost per unit transported from farmer i to sorting center j ,

c_{jd} is the cost per unit transported from sorting center j to distribution center d ,

c_{dm} is the cost per unit transported from distribution center d to market m ,

c_{dn} is the cost per unit transported from distribution center d to processing industry n ,

c_{nv} is the cost per unit transported from processing industry n to valorization industry v

5.6. Constraints

Flow Conservation:

$$\begin{aligned} \sum_j X_{ij} &= P_i, & \forall i \in I, & \text{ where } P_i \text{ is production quantity of farmer } i \\ \sum_d Y_{jd} &= \sum_i X_{ij}, & \forall j \in J \\ \sum_m Z_{dm} + \sum_n W_{dn} &= \sum_j Y_{jd}, & \forall d \in D \end{aligned}$$

Capacity Constraints:

$$\begin{aligned} X_{ij} &\leq \text{cap}_{ij}, & \forall i, j \\ Y_{jd} &\leq \text{cap}_{jd}, & \forall j, d \end{aligned}$$

where:

cap_{ij} : Capacity of transportation from farmer i to sorting center j

cap_{jd} : Capacity of transportation from sorting center j to distribution center d

Quality Constraints: Low-quality products are sent to processing industries. Rotten products are sent to valorization industries.

Demand Fulfillment:

$$\sum_d Z_{dm} \geq D_m, \quad \forall m \in M \quad \text{where } D_m \text{ is the market demand}$$

Non-negativity constraints:

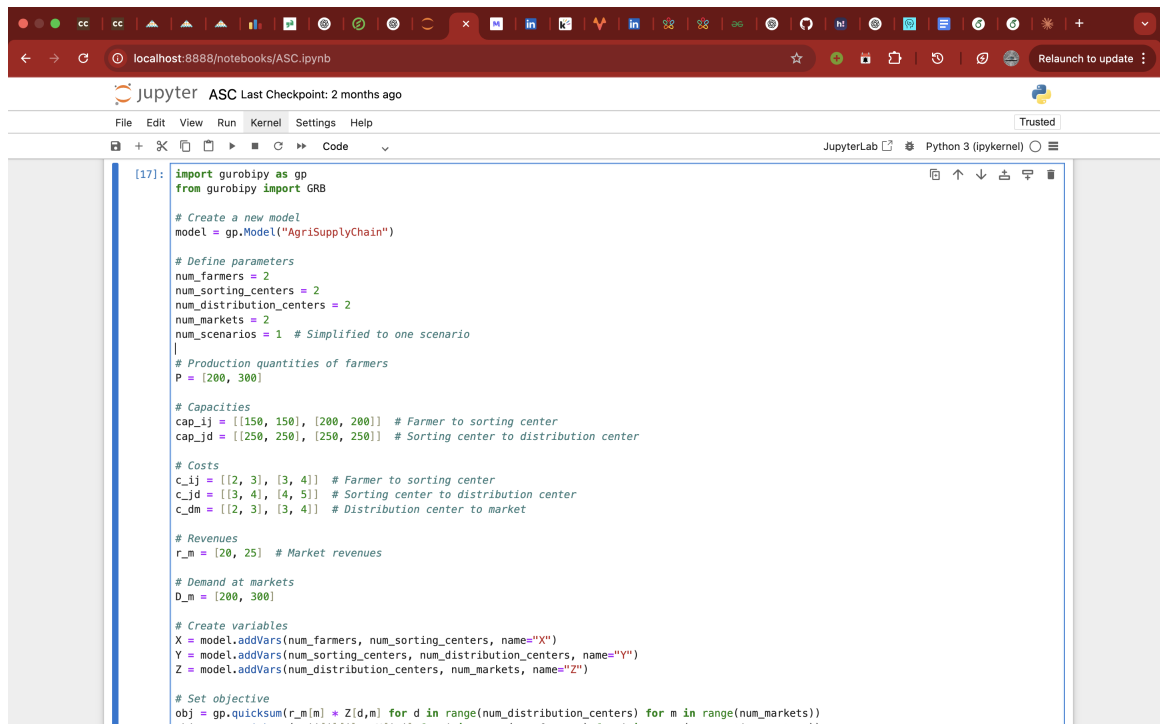
$$X_{ij}, Y_{jd}, Z_{dm}, W_{dn}, U_{nv} \geq 0, \quad \forall i, j, d, m, n, v$$

CHAPTER 6

6. IMPLEMENTATION OF AGRICULTURAL SUPPLY CHAIN OPTIMIZATION MODEL

6.1. Model Implementation in Python

To solve the agricultural supply chain optimization problem, we implemented the mathematical model using the Gurobi optimization solver through its Python interface, `gurobipy`. The implementation includes the definition of parameters, decision variables, objective function, and constraints as described in the mathematical formulation.



```
[17]: import gurobipy as gp
      from gurobipy import GRB

      # Create a new model
      model = gp.Model("AgriSupplyChain")

      # Define parameters
      num_farmers = 2
      num_sorting_centers = 2
      num_distribution_centers = 2
      num_markets = 2
      num_scenarios = 1 # Simplified to one scenario
      |
      # Production quantities of farmers
      P = [200, 300]

      # Capacities
      cap_ij = [[150, 150], [200, 200]] # Farmer to sorting center
      cap_jd = [[250, 250], [250, 250]] # Sorting center to distribution center

      # Costs
      c_ij = [[2, 3], [3, 4]] # Farmer to sorting center
      c_jd = [[3, 4], [4, 5]] # Sorting center to distribution center
      c_dm = [[2, 3], [3, 4]] # Distribution center to market

      # Revenues
      r_m = [20, 25] # Market revenues

      # Demand at markets
      D_m = [200, 300]

      # Create variables
      X = model.addVars(num_farmers, num_sorting_centers, name="X")
      Y = model.addVars(num_sorting_centers, num_distribution_centers, name="Y")
      Z = model.addVars(num_distribution_centers, num_markets, name="Z")

      # Set objective
      obj = gp.quicksum(r_m[m] * Z[d,m] for d in range(num_distribution_centers) for m in range(num_markets))
      obj += gp.quicksum(c_ij[i,j] * X[i,j] for i in range(num_farmers) for j in range(num_sorting_centers))
```

Figure 6.1: 1 Python code for the agricultural supply chain optimization model

The code above demonstrates the following key components:

1. Model initialization and parameter definition
2. Creation of decision variables
3. Formulation of the objective function

```

# Set objective
obj = gp.quicksum(r[m] * Z[d,m] for d in range(num_distribution_centers) for m in range(num_markets))
obj += gp.quicksum(c_ij[i][j] * X[i,j] for i in range(num_farmers) for j in range(num_sorting_centers))
obj += gp.quicksum(c_jd[j][d] * Y[j,d] for j in range(num_sorting_centers) for d in range(num_distribution_centers))
obj += gp.quicksum(c_dm[d][m] * Z[d,m] for d in range(num_distribution_centers) for m in range(num_markets))

model.setObjective(obj, GRB.MAXIMIZE)

# Add constraints
# Flow conservation constraints
for i in range(num_farmers):
    model.addConstr(gp.quicksum(X[i,j] for j in range(num_sorting_centers)) <= P[i])

for j in range(num_sorting_centers):
    model.addConstr(gp.quicksum(Y[j,d] for d in range(num_distribution_centers)) ==
                    gp.quicksum(X[i,j] for i in range(num_farmers)))

for d in range(num_distribution_centers):
    model.addConstr(gp.quicksum(Z[d,m] for m in range(num_markets)) ==
                    gp.quicksum(Y[j,d] for j in range(num_sorting_centers)))

# Capacity constraints
for i in range(num_farmers):
    for j in range(num_sorting_centers):
        model.addConstr(X[i,j] <= cap_ij[i][j])

for j in range(num_sorting_centers):
    for d in range(num_distribution_centers):
        model.addConstr(Y[j,d] <= cap_jd[j][d])

# Demand fulfillment constraints
for m in range(num_markets):
    model.addConstr(gp.quicksum(Z[d,m] for d in range(num_distribution_centers)) >= D_m[m])

# Optimize model
model.optimize()

# Print results

```

Figure 6.2: 2 Python code for the agricultural supply chain optimization model

```

# Print results
if model.status == GRB.OPTIMAL:
    print('\nOptimal objective: %g' % model.objVal)
    print('\nOptimal values:')
    for v in model.getVars():
        if v.x > 1e-6: # only print non-zero values
            print('%s %g' % (v.varName, v.x))
else:
    print('No optimal solution found')
    print('Optimization status:', model.status)

```

Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (mac64[arm] -- Darwin 22.6.0 22G74)

CPU model: Apple M1
Thread count: 8 physical cores, 8 logical processors, using up to 8 threads

Optimize a model with 16 rows, 12 columns and 32 nonzeros
Model fingerprint: 0xb8ccd2bd
Coefficient statistics:
Matrix range [1e+00, 1e+00]
Objective range [2e+00, 2e+01]
Bounds range [0e+00, 0e+00]
RHS range [2e+02, 3e+02]
Presolve removed 8 rows and 0 columns
Presolve time: 0.01s
Presolved: 8 rows, 12 columns, 24 nonzeros

Iteration	Objective	Primal Inf.	Dual Inf.	Time
0	3.9839078e+04	2.502505e+02	0.000000e+00	0s
8	6.9000000e+03	0.000000e+00	0.000000e+00	0s

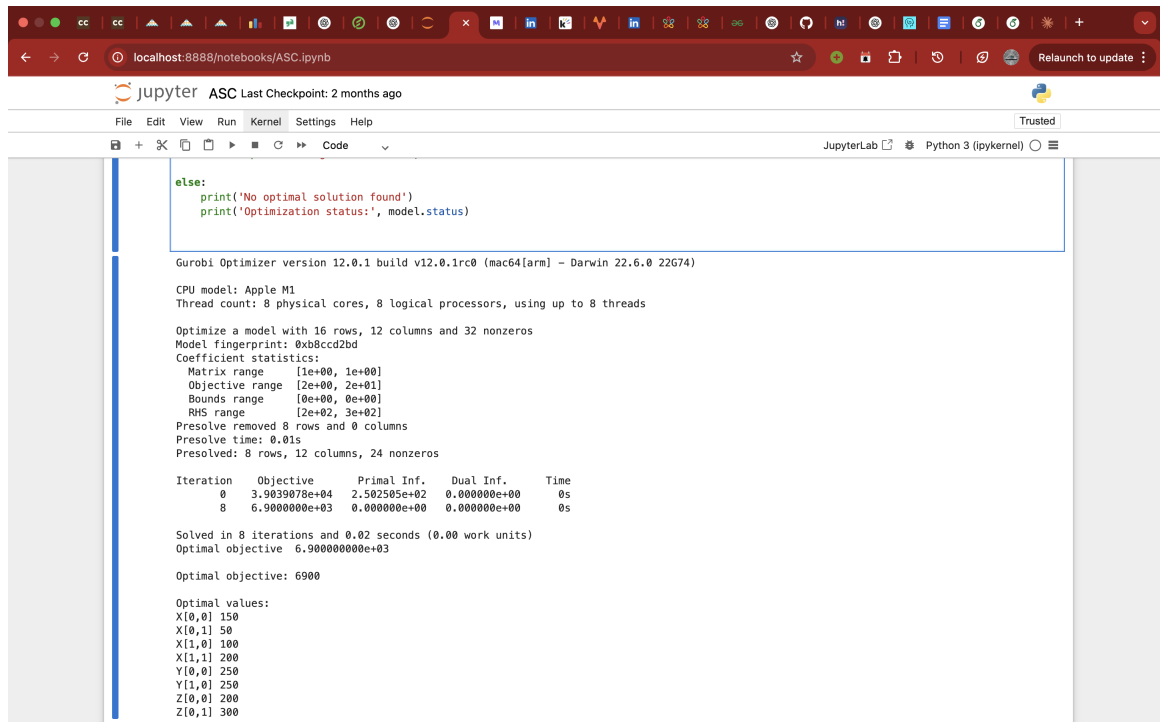
Solved in 8 iterations and 0.02 seconds (0.00 work units)
Optimal objective 6.900000000e+03
Optimal objective: 6900
Optimal values:
X[0,0] 150

Figure 6.3: 3 Python code for the agricultural supply chain optimization model

4. Implementation of constraints
5. Model optimization and result retrieval

6.2. Optimization Results

After running the optimization model with the given parameters, we obtained the following results:



The screenshot shows a JupyterLab window with a code editor and a terminal output. The code in the editor is a simple Python snippet that prints a message if no optimal solution is found and prints the optimization status otherwise. The terminal output shows the Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (mac64[arm] - Darwin 22.6.0 22G74) running on an Apple M1 CPU. It reports that the model was solved in 8 iterations and 0.02 seconds, achieving an optimal objective value of 6900. The optimal values for the decision variables are listed as follows:

```
Iteration   Objective   Primal Inf.   Dual Inf.   Time
0           3.9039078e+04  2.502505e+02  0.000000e+00  0s
8           6.9000000e+03  0.000000e+00  0.000000e+00  0s
```

Solved in 8 iterations and 0.02 seconds (0.00 work units)
Optimal objective 6.900000000e+03
Optimal objective: 6900
Optimal values:
X[0,0] 150
X[0,1] 50
X[1,0] 100
X[1,1] 200
Y[0,0] 250
Y[1,0] 250
Z[0,0] 200
Z[0,1] 300

Figure 6.4: Optimization results showing optimal objective value and decision variables

The optimization results provide insights into:

1. The optimal objective value, representing the maximum profit achieved
2. Optimal flow quantities between different entities in the supply chain
3. Utilization of capacities at various stages of the supply chain

7. FUTURE ASPECTS

These results demonstrate the effectiveness of the optimization model in determining the most efficient flow of products through the agricultural supply chain, balancing costs, revenues, and constraints. The creation and deployment of the digital agricultural value chain platform have tremendous potential in revolutionizing the fresh produce market, especially by overcoming existing inefficiencies in pricing, inventory management, and quality control. Nevertheless, for the platform to scale effectively, it is crucial to evaluate and plan for its financial and operational viability, create a comprehensive system, and conduct pilot testing to validate its effectiveness. The following section discusses these future elements, outlining steps required for effective implementation and growth.

7.1. Complete System Design and Prototype Development

The development of a fully functional system requires detailed planning and the creation of a working prototype. The key components of the system include:

- **Platform Architecture:** The system should have a cloud-based architecture with microservices, allowing for scalability and flexibility. Key components include:
 - **Inventory Management:** Real-time tracking of inventory and demand forecasting.
 - **Quality Control:** IoT sensors and AI for real-time grading and quality assessment.
 - **Pricing and Payment:** A dynamic pricing model integrated with secure payment gateways for seamless transactions.
- **Prototype:** The prototype will showcase a basic version of the platform with key features:
 - A web and mobile app for farmers and grocers to interact with the platform.
 - Basic functionalities such as product listing, price discovery, and order management.

CHAPTER 8

8. CONCLUSION

The prospects of the digital agricultural value chain platform are bright, with huge potential to make operations smoother, enhance quality control, and provide equitable pricing. Its success and scalability will, however, depend on rigorous financial and operational feasibility studies, the creation of an integrated system, pilot runs with stakeholders, and strategic growth based on feedback. Through prudent planning and implementation, this platform has the potential to revolutionize the agricultural industry, supporting sustainable methods, improving farmers' revenues, and enhancing consumer satisfaction.

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